

EDUCATION

University of [redacted] GPA 3.8/4.0
B.S. in Computer Science & B.S. in Computer Engineering Expected May 2027

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Machine Organization, Machine Learning, Linear Algebra, Big Data.
- Honors/Awards: **1st Place [redacted] Hackathon** (200+ participants) and Best Presentation; **Top 3** in University Madness Startup Pitch Competition and Audience Favorite (9 Universities, 25+ Teams), **Top 3** in [redacted] Launchpad Entrepreneurship Challenge, Dean's Honors List x2.

EXPERIENCE

[redacted] - Algorithmic Systems & Automation Software Engineer (Intern) May 2025 – Aug 2025

- Developed a copy-trading engine in C#/.NET (DI, async/await), implementing real-time data pipelines, stream processing, and automated data logging with **≤10 ms latency** across multiple accounts. Created an interactive copy-trading UI with **Dockable + DataGrid** in **WPF/XAML** with error/connection monitoring.
- Built an **XGBoost** algorithm (feature engineering, k-fold CV/early stopping), deployed via **REST API** integrated with .NET backend to drive real-time parameter tuning, automating **90%** of algorithm feature updates and reducing manual tuning by **70%**.

The Computational Optics Group - Machine Learning Researcher Jan 2025 – Apr 2025

- Improved NLOS optical sensor SNR by **15 dB** by building CNN-based denoising pipelines in **MATLAB** with **GPU acceleration** (CUDA/gpuArray), vectorized ops, batch inference, and parallel computing.
- Automated parameter tuning via **Bayesian optimization** for 3D reconstruction; implemented cross-validation, PSNR/SSIM logging, and runtime/memory profiling to accelerate CNN pipeline and establish scalability baseline.

[redacted] - Machine Learning Engineer (Intern) Aug 2022 – Jun 2023

- Reduced scientific abstract creation time by **90%** by productionizing a Python **NLP** pipeline with fine-tuned **transformers** (BERT, GPT, T5), with tokenization, preprocessing, and quantized/batched **PyTorch** inference
- Built an **ETL pipeline** for a web-scraped training dataset with data cleaning, validation, and normalization.
- Deployed a **Python-CGI** web app with file upload and JSON output; implemented logging, exception handling, and usage analytics, and integrated input validation and error reporting/debugging to support researchers.

PROJECTS

[redacted] | [redacted].ai

- Co-founded a **4x award-winning AI tool** that slashes home insurance claim prep from 20+ hours to minutes.
- Enhanced **YOLOv11** with depth estimation, spatial context modeling, attention mechanisms, and image augmentation modules for enhanced multi-object detection and 3D scene understanding.
- Built a Python-based multimodal **RAG** pipeline integrating **CLIP**-based embedding retrieval, vector similarity search, and **Gemini API** for semantic image understanding, context-aware reasoning, and automated valuation.
- Deployed on **AWS ECS** with **Docker** and **Supabase** for authentication, optimized backend to support **5,000+** users, processing **22,000+** images, with over **\$20 million** worth of home assets.

[redacted] | [redacted].dev

- Created **Firestore** backend to manage user auth, session storage, and real-time mobile data sync across sessions.
- Built a relational **SQL** schema to structure quizzes, scoring rules, and adaptive content for financial literacy app.

Multi-Environment Reinforcement Learning Agent

- Designed **Q-learning** agent with function approximation, epsilon decay, reward shaping, and experience replay.
- Implemented a **genetic algorithm** to optimize learning rate, exploration, and discount factors, enabling automated tuning and improved policy convergence, achieving performance in multiple OpenAI Gym environments

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, MATLAB

Frameworks/Libraries: .NET, Tensorflow, Flask, React, Pandas, NumPy, Keras, Sklearn, Matplotlib, OpenCV

Tools: GitHub, AWS, Docker, Firebase, Supabase, Python-CGI, Linux, Google Cloud, REST APIs, NinjaScript